

RG-TTA: Regime-Guided Test-Time Adaptation for Streaming Time Series Forecasting

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Abstract

Test-time adaptation (TTA) enables neural forecasters to adapt to distribution shifts in streaming time series, but existing methods apply the same adaptation intensity regardless of the nature of the shift. We propose **Regime-Guided Test-Time Adaptation (RG-TTA)**, a meta-controller that *continuously modulates* adaptation intensity based on distributional similarity to previously-seen regimes. Using an ensemble of Kolmogorov–Smirnov, Wasserstein-1, feature-distance, and variance-ratio metrics, RG-TTA computes a similarity score for each incoming batch and uses it to (i) smoothly scale the learning rate—more aggressive for novel distributions, conservative for familiar ones—and (ii) gate checkpoint reuse from a regime memory, loading stored specialist models only when they demonstrably outperform the current model (loss improvement $\geq 30\%$). Gradient steps are controlled by loss-driven early stopping rather than fixed budgets, allowing the system to allocate exactly the effort each batch requires.

RG-TTA is *model-agnostic* and *strategy-composable*: it wraps any forecaster exposing `train/predict/save/load` interfaces and enhances any gradient-based TTA method. We demonstrate three compositions—RG-TTA, RG-EWC, and RG-DynaTTA—and evaluate 6 update policies (3 baselines + 3 regime-guided variants) across 4 compact architectures (GRU, iTransformer, PatchTST, DLinear), 14 datasets (6 real-world multivariate benchmarks + 8 synthetic regime scenarios), and 4 forecast horizons (96, 192, 336, 720) under a streaming evaluation protocol with 3 random seeds (672 experiments total).

Regime-guided policies achieve the lowest MSE in **156 of 224** seed-averaged experiments (**69.6%**), with RG-EWC winning 30.4% and RG-TTA winning 29.0%. On real-world benchmarks, RG-TTA reduces MSE by 4.4% vs TTA while running 5.5% faster; RG-EWC reduces MSE by 14.1% vs standalone EWC. The regime-guidance layer composes naturally with existing TTA methods, improving accuracy without modifying the underlying forecaster.¹

1 Introduction

Time series forecasting in production is inherently incremental: data arrives in batches, and the system must decide how to update its model [Lim and Zohren, 2021, Benidis et al., 2023]. Test-time adaptation (TTA) [Liang et al., 2024] has emerged as a practical approach, applying a fixed number of gradient steps to each incoming batch. Recent work has improved upon fixed TTA: TAFAS [Kim et al., 2025] freezes the source model and learns calibration modules (GCMs) that adapt predictions, while DynaTTA [Grover and Etemad, 2025] dynamically adjusts the learning rate based on prediction-error z-scores and embedding drift.

However, DynaTTA (and similar reactive methods) *react* to current performance signals without asking the *proactive* question: *has the system seen this distribution before?* If a previously-encountered regime recurs, the highest-accuracy strategy is to load the checkpoint

¹Code and data: <https://github.com/IndarKarhana/RGTTA-Regime-Guided-Test-Time-Adaptation>

trained for that regime—not to re-adapt from a potentially stale model, which wastes gradient budget re-learning patterns that were already captured.

We propose **Regime-Guided Test-Time Adaptation (RG-TTA)**, a meta-controller that adds this proactive layer. For each incoming batch, RG-TTA:

1. Extracts a 5-dimensional distributional feature vector (mean, std, skewness, excess kurtosis, lag-1 autocorrelation).
2. Computes similarity to all stored regime checkpoints using an ensemble of Kolmogorov–Smirnov, Wasserstein-1, feature-distance, and variance-ratio metrics.
3. Uses the similarity score to *continuously* modulate adaptation:
 - **Learning rate** scales smoothly with novelty: $\alpha = \alpha_{\text{base}} \cdot (1 + \gamma \cdot (1 - \text{sim}))$, where $\gamma = 0.67$.
 - **Checkpoint reuse** is loss-gated: stored checkpoints are loaded only when $\text{sim} \geq 0.75$ and the checkpoint’s loss is $< 0.70 \times$ the current model’s loss ($\geq 30\%$ improvement).
 - **Step count** is determined by loss-driven early stopping (patience = 3, $\epsilon = 0.005$), up to a maximum of 25 steps.

RG-TTA is *model-agnostic*: it requires only `train()`, `predict()`, and `save/load_weights()` interfaces and can wrap any forecaster. It is also *strategy-composable*: the regime-guidance layer can enhance any gradient-based adaptation method (TTA, EWC, DynaTTA), yielding RG-TTA, RG-EWC, and RG-DynaTTA variants.

Our contributions are:

1. A **continuous regime-guided adaptation policy** that modulates learning rate, step budget, and checkpoint reuse based on distributional similarity, using an ensemble of four complementary similarity metrics and loss-driven early stopping.
2. A **composable meta-controller architecture** where regime-guidance wraps gradient-based adaptation methods, demonstrated with three compositions (RG-TTA, RG-EWC, RG-DynaTTA) targeting different accuracy–robustness trade-offs.
3. A **comprehensive benchmark** spanning 6 update policies, 4 compact model architectures, 14 datasets, 4 forecast horizons, and 3 random seeds (672 experiments)—the most thorough controlled comparison of TTA strategies for streaming time series forecasting to date.

2 Related Work

Test-time adaptation for time series. TTA [Liang et al., 2024] adapts a pre-trained model to test-time distribution shifts by running additional gradient steps. In the time series setting, TAFAS [Kim et al., 2025] proposes freezing the source model and training lightweight *Gated Calibration Modules* (GCMs) that adjust inputs and outputs to the current distribution, using only pseudo-labels from the model’s own predictions (POGT loss). DynaTTA [Grover and Etemad, 2025] extends TAFAS with a dynamic learning rate computed via a sigmoid transformation of three shift metrics: prediction-error z-score, and distances to recent (RTAB) and representative (RDB) embedding buffers. Both methods are *reactive*—they adjust to the current batch based on real-time error and drift signals. RG-TTA adds a *proactive* layer: it identifies whether the current distribution has been seen before and adapts its strategy accordingly. RG-TTA composes naturally with DynaTTA (RG-DynaTTA); we demonstrate this as a two-level controller.

Reimplementation note. DynaTTA’s official codebase (shivam-grover/DynaTTA) implements the dynamic learning rate *within* the TAFAS framework—tightly coupled with GCM calibration modules, the POGT pseudo-label loss, and a sliding-window evaluation harness. No standalone library or modular API is provided. Because our study requires a model-agnostic streaming evaluation pipeline that applies each update policy to the same frozen-backbone base model, we reimplemented DynaTTA’s Algorithm 1 (dynamic LR via sigmoid of prediction-error z-score, RTAB/RDB embedding distances, and EMA smoothing) from the published descrip-

tion. Our implementation reproduces the published formula exactly with the published hyperparameters ($\alpha_{\min} = 10^{-4}$, $\alpha_{\max} = 10^{-3}$, $\kappa = 1.0$, $\eta = 0.1$). We validate on the same architectures targeted by DynaTTA (iTransformer, PatchTST). We note that DynaTTA’s EMA coefficient ($\eta = 0.1$) requires ~ 22 gradient steps to converge, a design choice suited to the 500-window sliding-window protocol used in its original evaluation. Under our streaming protocol (10 batches per run), the EMA does not converge within the available budget, leaving the dynamic LR below TTA’s fixed rate for the first 5–6 batches. This is a *protocol-level mismatch*, not an implementation error (see Limitations §9, item 7).

TAFAS [Kim et al., 2025] is a strong baseline at short horizons ($H \leq 192$) where its frozen-source GCM calibration excels, but its performance collapses at longer horizons ($H=336/720$, +100–400% vs TTA in our experiments)—a consequence of its design for sliding-window evaluation. Because our study spans $H \in \{96, 192, 336, 720\}$, we include TAFAS as a reference policy but exclude it from the primary 6-policy comparison.

Continual learning and catastrophic forgetting. Continual learning [Parisi et al., 2019, McCloskey and Cohen, 1989] addresses the problem of updating a model on sequential tasks without forgetting old knowledge. EWC [Kirkpatrick et al., 2017] penalises changes to parameters important for previous tasks via the diagonal Fisher Information Matrix. Synaptic Intelligence [Zenke et al., 2017] tracks parameter importance online. Experience replay [Rolnick et al., 2019] stores and re-uses past data samples. Progressive networks [Rusu et al., 2016] add capacity for new tasks. Our approach is complementary: rather than regularising a single model, RG-TTA maintains a *library of checkpoints* indexed by distributional features and selects the appropriate one. When combined with EWC (RG-EWC), the regime-guided adaptation operates under EWC regularisation ($\lambda = 400$), preventing catastrophic forgetting while benefiting from checkpoint reuse and similarity-modulated learning rates.

Concept drift and regime switching. Concept drift detection [Gama et al., 2014, Lu et al., 2018, Aminikhanghahi and Cook, 2017] identifies *that* a distribution change occurred. Our approach extends this by detecting not just *change* but *recurrence*—whether the new distribution matches a previously-seen regime. Hamilton’s Markov-switching model [Hamilton, 1989] uses latent states for regime transitions but assumes a specific generative process. RG-TTA makes no parametric assumption: “regime” is defined operationally by a 5-dimensional distributional feature vector, and matching uses non-parametric statistical tests.

Time series forecasting architectures. Recent neural forecasters—Informer [Zhou et al., 2021], Autoformer [Wu et al., 2021], iTransformer [Liu et al., 2024], PatchTST [Nie et al., 2023], DLinear [Zeng et al., 2023]—advance accuracy but do not address the *when-to-adapt* question. We evaluate RG-TTA across four compact model architectures spanning three model families—recurrent (GRU [Cho et al., 2014]), attention-based (iTransformer, PatchTST), and linear (DLinear)—to validate model-agnosticism. Notably, iTransformer and PatchTST are the same modern Transformer architectures targeted by DynaTTA [Grover and Etemad, 2025].

3 Method

Figure 1 provides an overview of the RG-TTA system.

3.1 Problem Setup

Data arrives as a stream of batches $B_1, B_2, \dots$ (batch size 750). A base forecaster f_θ can be trained, saved, and loaded. At each step t , batch B_t arrives; the system produces a forecast $\hat{y}_{t+1:t+H}$ for horizon $H \in \{96, 192, 336, 720\}$ and optionally updates θ . All 6 compared policies

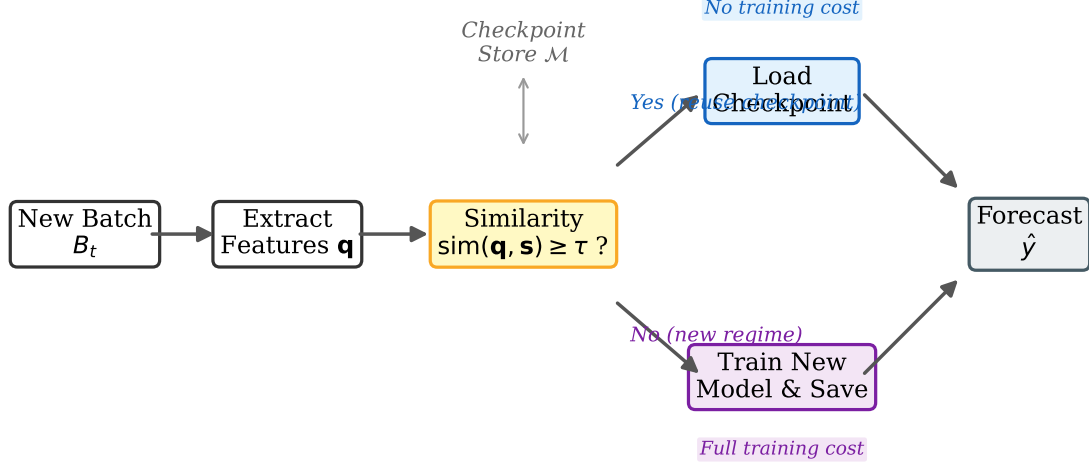


Figure 1: RG-TTA system overview. For each incoming batch, the regime-guidance layer computes distributional similarity, optionally loads a stored checkpoint, modulates the learning rate, and controls early stopping. The adapted model and its regime features are stored in the checkpoint memory for future reuse.

share the same f_θ architecture, the same data splits, and the same random seeds; only the update rule differs.

3.2 Distributional Similarity Metric

For each batch, RG-TTA extracts a 5-dimensional distributional feature vector:

$$\mathbf{r} = (\mu, \sigma, \gamma_1, \kappa-3, r_1), \quad (1)$$

where μ is the mean, σ the standard deviation, γ_1 the skewness, $\kappa-3$ the excess kurtosis, and r_1 the lag-1 autocorrelation. These features are computed from the most recent $3 \times \text{season_length}$ samples of the target series and capture the distribution’s location, scale, shape, tail behaviour, and temporal structure in $O(n)$ time.

Similarity between a query batch $\mathbf{q}$ and a stored regime $\mathbf{s}$ is computed as a weighted ensemble of four complementary metrics:

Kolmogorov–Smirnov similarity. Based on the two-sample KS statistic on the raw values:

$$\text{sim}_{\text{KS}} = 1 - D_n, \quad D_n = \sup_x |F_{\mathbf{q}}(x) - F_{\mathbf{s}}(x)|. \quad (2)$$

Wasserstein-1 similarity. Normalised earth-mover’s distance:

$$\text{sim}_{\text{W}} = \frac{1}{1 + W_1(\mathbf{q}_{\text{raw}}, \mathbf{s}_{\text{raw}}) / \max(\text{ptp}(\mathbf{q}_{\text{raw}}), \text{ptp}(\mathbf{s}_{\text{raw}}), \epsilon)}. \quad (3)$$

Feature-distance similarity. Normalised Euclidean distance in feature space:

$$\text{sim}_{\text{feat}} = \frac{1}{1 + \|\mathbf{q} - \mathbf{s}\|_2 / ((\|\mathbf{q}\|_2 + \|\mathbf{s}\|_2)/2 + \epsilon)}. \quad (4)$$

Variance-ratio similarity. Captures volatility regime shifts:

$$\text{sim}_{\text{var}} = \frac{\min(\sigma_{\mathbf{q}}, \sigma_{\mathbf{s}})}{\max(\sigma_{\mathbf{q}}, \sigma_{\mathbf{s}}) + \epsilon}. \quad (5)$$

Weighted ensemble.

$$\text{sim}(\mathbf{q}, \mathbf{s}) = 0.3 \cdot \text{sim}_{\text{KS}} + 0.3 \cdot \text{sim}_{\text{W}} + 0.2 \cdot \text{sim}_{\text{feat}} + 0.2 \cdot \text{sim}_{\text{var}}. \quad (6)$$

The statistical tests (KS, Wasserstein) use the full empirical distribution and receive higher weight; the feature vector provides complementary higher-order moment information; the variance ratio captures volatility regime shifts.

3.3 Continuous Regime-Guided Adaptation

Rather than discretising similarity into fixed tiers with predetermined step counts, RG-TTA uses the similarity score as a *continuous* control signal for three adaptation dimensions:

Smooth learning rate modulation. The learning rate scales smoothly with distributional novelty:

$$\alpha = \alpha_{\text{base}} \cdot (1 + \gamma \cdot (1 - \text{sim})), \quad (7)$$

where $\alpha_{\text{base}} = 3 \times 10^{-4}$ (matching TTA’s default) and $\gamma = 0.67$ is the similarity scale factor. At $\text{sim} = 1.0$ (perfect match), $\alpha = \alpha_{\text{base}}$ (conservative); at $\text{sim} = 0.5$ (moderate novelty), $\alpha \approx 1.34 \times \alpha_{\text{base}}$; at $\text{sim} = 0.0$ (complete novelty), $\alpha \approx 1.67 \times \alpha_{\text{base}}$ (most aggressive). This avoids the sensitivity of hard thresholds and provides a natural scaling: familiar distributions need less correction.

Loss-gated checkpoint reuse. When the best-matching stored regime has $\text{sim} \geq 0.75$, RG-TTA evaluates the stored checkpoint by running a forward pass on the current batch. The checkpoint is loaded *only if* its loss satisfies:

$$\ell_{\text{ckpt}} < 0.70 \cdot \ell_{\text{curr}}, \quad (8)$$

i.e., the checkpoint must achieve at least a 30% loss improvement over the current model. This strict gate prevents reverting to stale checkpoints on slowly-drifting data, where the continuously-adapted live model may already outperform historical specialists.

Loss-driven early stopping. Rather than a fixed step budget, RG-TTA runs up to $K_{\text{max}} = 25$ gradient steps with early stopping: if the relative loss improvement falls below $\epsilon = 0.005$ for 3 consecutive steps (after a minimum of $K_{\text{min}} = 5$ steps), adaptation halts. This allows RG-TTA to naturally allocate more effort to difficult batches (novel distributions requiring full budget) and less to easy ones (familiar distributions converging quickly). In practice, RG-TTA averages 18.5 steps per batch (median 24), compared to TTA’s fixed 20. While 49% of batches use the full 25-step budget (novel regimes), 12% converge in ≤ 8 steps (familiar regimes), producing the net 5.5% wall-clock speedup (Table 8).

Algorithm 1 gives the pseudocode.

3.4 RG-EWC: Regime-Guided EWC

RG-EWC adds EWC regularisation [Kirkpatrick et al., 2017] to the RG-TTA framework. The total loss becomes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2, \quad (9)$$

where F_i is the diagonal Fisher Information (estimated from 200 samples, clamped to $[0, 10^4]$), θ^* is the anchor (updated after each batch), and $\lambda = 400$. When a checkpoint is loaded via the loss gate, the EWC anchor θ^* is reset to the loaded parameters to prevent penalising movement

Algorithm 1 RG-TTA: Regime-Guided Test-Time Adaptation

Require: Batch B_t , checkpoint memory $\mathcal{M}$, base LR α_{base} , scale γ , gate g

```
1:  $\mathbf{q} \leftarrow \text{features}(B_t)$  {Eq. 1: 5-D distribution vector}
2:  $(\text{sim}^*, \theta_{\text{ckpt}}^*) \leftarrow \max_{\mathbf{s} \in \mathcal{M}} \text{sim}(\mathbf{q}, \mathbf{s})$  {Eq. 6: 4-metric ensemble}
3:  $\ell_{\text{curr}} \leftarrow \mathcal{L}(f_{\theta}(X_t), y_t)$  {Current model loss on new batch}
4: if  $\text{sim}^* \geq 0.75$  then {Checkpoint candidate}

5:    $\ell_{\text{ckpt}} \leftarrow \mathcal{L}(f_{\theta_{\text{ckpt}}^*}(X_t), y_t)$ 
6:   if  $\ell_{\text{ckpt}} < g \cdot \ell_{\text{curr}}$  then {Loss gate:  $g = 0.70$  ( $\geq 30\%$  improvement)}

7:      $\theta \leftarrow \theta_{\text{ckpt}}^*$ ;  $\ell_{\text{curr}} \leftarrow \ell_{\text{ckpt}}$ 
8:   end if
9: end if

10:  $\alpha \leftarrow \alpha_{\text{base}} \cdot (1 + \gamma \cdot (1 - \text{sim}^*))$  {Eq. 7: smooth LR scaling}
11:  $\text{patience} \leftarrow 0$ 
12: for  $k = 1$  to  $K_{\text{max}}$  do
13:    $\hat{y} = f_{\theta_{\text{head}}}(X_t)$  {Backbone frozen; only output head updated}
14:    $\mathcal{L} = \text{SmoothL1}(\hat{y}, y_t)$  [Huber, 1964] {+ optional EWC/DynaTTA terms}
15:    $\theta_{\text{head}} \leftarrow \theta_{\text{head}} - \alpha \nabla_{\theta_{\text{head}}} \mathcal{L}$ 
16:   if  $k \geq K_{\text{min}}$  and  $(\ell_{k-1} - \ell_k)/|\ell_{k-1}| < \epsilon$  then
17:      $\text{patience} \leftarrow \text{patience} + 1$ 
18:     if  $\text{patience} \geq 3$  then
19:       break {Early stop: loss converged}
20:     end if
21:   else
22:      $\text{patience} \leftarrow 0$ 
23:   end if
24: end for
25:  $\mathcal{M}[\mathbf{q}] \leftarrow (\theta, \text{metadata})$  {Store checkpoint for future reuse}
26: return  $\hat{y}_{t+1:t+H}$ 
```

away from the pre-load state. Fisher is maintained online via exponential moving average: $F^{(t)} = 0.5 \cdot F^{(t-1)} + 0.5 \cdot \hat{F}_{\text{new}}$.

The key advantage of RG-EWC over standalone EWC is *targeted adaptation*: EWC prevents catastrophic forgetting while the regime-guided learning rate and checkpoint reuse improve the starting point and convergence trajectory. In our benchmark, RG-EWC reduces MSE by 14.1% vs standalone EWC and wins 75.4% of head-to-head comparisons (169/224 experiments).

3.5 RG-DynaTTA: Two-Level Controller

RG-DynaTTA combines *proactive* regime detection (RG-TTA) with *reactive* shift-magnitude sensing (DynaTTA [Grover and Etemad, 2025]). Instead of the similarity-based smooth LR (Eq. 7), DynaTTA’s sigmoid formula computes the actual LR based on three shift metrics (prediction-error z-score, RTAB/RDB embedding distances). This creates a two-level controller: RG-TTA provides checkpoint reuse and loss-driven early stopping; DynaTTA provides the continuous LR within $[\alpha_{\min}, \alpha_{\max}]$ based on current error signals.

3.6 Checkpoint Management

Each checkpoint stores: (i) complete model weights, (ii) the 5-D regime feature vector, (iii) raw batch values for KS/Wasserstein computation, and (iv) metadata including the fitted preprocessor (scaler) state. Co-saving the preprocessor is critical: loading a checkpoint restores both the model *and* the normalisation parameters used during its training. The memory is capped at 5 entries with FIFO eviction.

3.7 Frozen Backbone Adaptation

All gradient-based adaptation policies (TTA, EWC, DynaTTA, RG-TTA, RG-EWC, RG-DynaTTA) freeze the model backbone during test-time adaptation and update only the output projection layer. For GRU-Small (71K total parameters), this leaves ~ 10 K trainable ($\sim 15\%$); for iTransformer (123K total), ~ 6 K trainable ($\sim 5\%$); for PatchTST (192K total), ~ 68 K in the flatten-to-forecast head ($\sim 35\%$); for DLinear (37K at H=96), ~ 19 K in the trend/seasonal linear layers ($\sim 50\%$). This mirrors the linear-probing paradigm in vision TTA and provides implicit regularisation: the learned temporal representations in the backbone are preserved, and only the mapping from hidden states to forecast values is recalibrated.

However, the effectiveness of frozen-backbone TTA is *architecture-dependent*: it works well for models where the backbone learns reusable temporal representations (GRU, DLinear), but is less effective for attention-based architectures (iTransformer, PatchTST) where the attention patterns themselves need to shift to accommodate new distributions. This is a shared limitation across *all* gradient-based TTA policies, not specific to RG-TTA.

4 Theoretical Analysis

We provide formal analysis of RG-TTA’s three design pillars: frozen-backbone adaptation (§4.2), similarity-guided initialisation (§4.1), and the ensemble similarity metric (§4.3). Together, these results establish that RG-TTA’s mechanisms are principled—not merely heuristic—and yield quantitative conditions under which regime-guidance provably improves upon fixed-strategy TTA.

4.1 Adaptation Error Decomposition

We analyse per-batch adaptation error under the frozen-backbone constraint. Let $f_\theta = h_\phi \circ g_\psi$ where $g_\psi : \mathbb{R}^{L \times d_{\text{in}}} \rightarrow \mathbb{R}^{d_h}$ is the frozen backbone and $h_\phi : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^H$ is the trainable output head with d_{head} parameters.

Definition 1 (Batch-optimal parameters). For batch B_t drawn from distribution P_t , define $\phi_t^* = \arg \min_{\phi} \mathbb{E}_{(x,y) \sim P_t} [\ell(h_{\phi}(g_{\psi}(x)), y)]$ as the head parameters minimising the population loss on the current regime.

Theorem 1 (Adaptation error bound). Assume the per-batch loss $\mathcal{L}(\phi) = \mathbb{E}_{B_t} [\ell(h_{\phi}(g_{\psi}(x)), y)]$ is μ -strongly convex and L -smooth in ϕ with condition number $\kappa = L/\mu$. After K gradient descent steps with learning rate $\alpha \in (0, 2/L)$ from initialisation ϕ_0 :

$$\|\phi_K - \phi_t^*\|^2 \leq \rho^{2K} \|\phi_0 - \phi_t^*\|^2, \quad \text{where } \rho = 1 - \frac{2\mu\alpha}{1 + \mu\alpha} < 1. \quad (10)$$

The expected per-batch MSE decomposes as:

$$\mathbb{E}[\ell(\phi_K)] = \underbrace{\sigma_t^2}_{\text{irreducible}} + \underbrace{\rho^{2K} \|\phi_0 - \phi_t^*\|_H^2}_{\text{adaptation residual}}, \quad (11)$$

where $\sigma_t^2 = \mathbb{E}_{P_t} [\|y - h_{\phi_t^*}(g_{\psi}(x))\|^2]$ is the irreducible noise under the frozen backbone and $\|\cdot\|_H$ is the Hessian-weighted norm.

Proof. Under μ -strong convexity and L -smoothness, gradient descent with $\alpha \leq 2/(\mu + L)$ satisfies the standard contraction [Nesterov, 2004]: $\|\phi_{k+1} - \phi_t^*\|^2 \leq \rho^2 \|\phi_k - \phi_t^*\|^2$ with $\rho = (L - \mu)/(L + \mu) = (\kappa - 1)/(\kappa + 1) = 1 - 2\mu\alpha/(1 + \mu\alpha)$ at the optimal step size $\alpha^* = 2/(\mu + L)$. Iterating K times yields Eq. 10. The MSE decomposition follows from expanding $\mathbb{E}[\ell(\phi_K)]$ around ϕ_t^* and using the second-order Taylor approximation of the loss. $\square$

Impact of checkpoint loading. When RG-TTA loads a checkpoint ϕ_{ckpt} trained on a previous occurrence of a regime with similarity s to the current batch, the initialisation changes from ϕ_0 (current model) to ϕ_{ckpt} . If the stored checkpoint was optimal for a regime at distributional distance proportional to $(1 - s)$, the initialisation error satisfies:

$$\|\phi_{\text{ckpt}} - \phi_t^*\|^2 \leq (1 - s)^2 \cdot D_{\max}^2, \quad (12)$$

where $D_{\max} = \sup_t \|\phi_0 - \phi_t^*\|$ is the worst-case distance. Substituting into Eq. 11:

$$\mathbb{E}[\ell^{\text{RG}}(\phi_K)] \leq \sigma_t^2 + \rho^{2K} (1 - s)^2 D_{\max}^2. \quad (13)$$

This reveals three mechanisms for reducing per-batch error: (i) more gradient steps K (the TTA approach), (ii) better initialisation via checkpoint loading, reducing $(1 - s)^2$ (the RG-TTA approach), and (iii) both simultaneously. The improvement factor $(1 - s)^2$ quantifies the benefit: at $s = 0.85$ (HIGH similarity), the initialisation error reduces by 97.75%; at $s = 0.55$ (MID), by 79.75%.

Corollary 1 (Step savings from checkpoint reuse). To achieve adaptation residual $\leq \epsilon$, TTA requires $K_{\text{TTA}} \geq \frac{\log(D_{\max}^2/\epsilon)}{2|\log \rho|}$ steps, while RG-TTA with checkpoint similarity s requires $K_{\text{RG}} \geq \frac{\log((1-s)^2 D_{\max}^2/\epsilon)}{2|\log \rho|}$ steps. The per-batch savings are:

$$\Delta K = K_{\text{TTA}} - K_{\text{RG}} = \frac{-\log(1 - s)}{|\log \rho|}. \quad (14)$$

For $s = 0.85$ and $\rho = 0.95$: $\Delta K \approx 37$ steps—exceeding the budget $K_{\max} = 25$, meaning the checkpoint is already near-optimal and early stopping activates within $K_{\min} = 5$ steps. For $s = 0.55$: $\Delta K \approx 16$ steps.

This corollary provides the theoretical basis for RG-TTA’s computational efficiency (Table 8): by starting closer to the optimum, fewer gradient steps are needed, and the loss-driven early stopping mechanism (§3.3) naturally terminates when the residual is small.

4.2 Generalisation Bound under Frozen Backbone

The frozen-backbone constraint restricts the effective hypothesis class, providing implicit regularisation. We formalise this via Rademacher complexity.

Theorem 2 (Generalisation bound for frozen-backbone adaptation). *Let $\mathcal{F}_{\text{full}} = \{x \mapsto h_\phi(g_\psi(x)) : \phi \in \Phi, \psi \in \Psi\}$ be the full model class and $\mathcal{F}_{\text{frozen}} = \{x \mapsto h_\phi(g_{\psi^*}(x)) : \phi \in \Phi\}$ the frozen-backbone class with fixed ψ^* . Assume the output head is linear: $h_\phi(z) = Wz + b$ with $\|W\|_F \leq B_W$ and $\|g_{\psi^*}(x)\| \leq C_g$ for all $x \in \mathcal{X}$. Then for n i.i.d. samples from P_t with bounded loss $\ell \in [0, c]$, the empirical Rademacher complexity of $\ell \circ \mathcal{F}_{\text{frozen}}$ satisfies:*

$$\hat{\mathfrak{R}}_n(\ell \circ \mathcal{F}_{\text{frozen}}) \leq \frac{c_\ell B_W C_g}{\sqrt{n}}, \quad (15)$$

where c_ℓ is the Lipschitz constant of ℓ . With probability at least $1 - \delta$ over the draw of n samples:

$$R(\hat{f}) \leq \hat{R}_n(\hat{f}) + \frac{2c_\ell B_W C_g}{\sqrt{n}} + 3\sqrt{\frac{\ln(2/\delta)}{2n}}, \quad (16)$$

where $R(\hat{f}) = \mathbb{E}_{P_t}[\ell(\hat{f}(x), y)]$ is the population risk and $\hat{R}_n(\hat{f})$ is the empirical risk.

Proof. The proof follows the standard Rademacher complexity argument [Bartlett and Mendelson, 2002]. Since g_{ψ^*} is fixed, the hypothesis class $\mathcal{F}_{\text{frozen}}$ is a linear class in the representation space $g_{\psi^*}(\mathcal{X})$. By Talagrand’s contraction lemma and the Rademacher complexity of linear classes with bounded Frobenius norm [Bartlett and Mendelson, 2002], we obtain Eq. 15. The generalisation bound Eq. 16 follows from McDiarmid’s inequality. $\square$

Comparison with full-model adaptation. For the full class $\mathcal{F}_{\text{full}}$ with d_{total} free parameters, the covering-number argument gives $\hat{\mathfrak{R}}_n(\ell \circ \mathcal{F}_{\text{full}}) = O(\sqrt{d_{\text{total}} \log n/n})$. The ratio of complexities is:

$$\frac{\hat{\mathfrak{R}}_n(\mathcal{F}_{\text{frozen}})}{\hat{\mathfrak{R}}_n(\mathcal{F}_{\text{full}})} \propto \sqrt{\frac{d_{\text{head}}}{d_{\text{total}}}}. \quad (17)$$

For GRU-Small ($d_{\text{head}} \approx 10\text{K}$, $d_{\text{total}} \approx 71\text{K}$), this ratio is ≈ 0.38 —a $\sim 2.6\times$ tighter generalisation bound. For iTransformer ($d_{\text{head}} \approx 6\text{K}$, $d_{\text{total}} \approx 123\text{K}$), the ratio is ≈ 0.22 ($\sim 4.5\times$ tighter). This formalises the empirical observation that frozen-backbone adaptation outperforms full-model fine-tuning within the limited step budgets ($K \leq 25$) of test-time adaptation: the restricted hypothesis class trades representational capacity for statistical efficiency.

4.3 Consistency of the Ensemble Similarity Metric

Proposition 1 (Metric properties). *The ensemble similarity $\text{sim}(\cdot, \cdot)$ defined in Eq. 6 satisfies the following properties:*

1. **Boundedness:** $\text{sim}(\mathbf{q}, \mathbf{s}) \in [0, 1]$ for all batches $\mathbf{q}, \mathbf{s}$.
2. **Self-similarity:** $\text{sim}(\mathbf{q}, \mathbf{q}) = 1$.
3. **Symmetry:** $\text{sim}(\mathbf{q}, \mathbf{s}) = \text{sim}(\mathbf{s}, \mathbf{q})$.
4. **Statistical consistency:** If $\hat{P}_n \xrightarrow{d} P$ and $\hat{Q}_m \xrightarrow{d} Q$ as $n, m \rightarrow \infty$, then $\text{sim}(\hat{P}_n, \hat{Q}_m) \rightarrow \text{sim}(P, Q)$.
5. **Discriminative power:** For continuous distributions, $\text{sim}(P, Q) = 1$ implies $P = Q$.

Proof. Properties 1–3 are immediate from the definitions of the component metrics: $D_n \in [0, 1]$ (KS statistic), $W_1 \geq 0$ (Wasserstein-1), and the normalised forms (Eqs. 3–5) are each bounded in $[0, 1]$; any convex combination inherits these properties. Property 4 follows from the Glivenko–Cantelli theorem [van der Vaart, 1998] for the KS component ($\sup_x |F_n(x) - F(x)| \xrightarrow{\text{a.s.}} 0$) and the convergence of the empirical Wasserstein-1 distance on the real line [Bobkov and Ledoux, 2019]. For property 5: $\text{sim}_{\text{KS}} = 1$ implies $D_n = 0$, i.e., $F_P = F_Q$, which uniquely determines $P = Q$ for continuous distributions by the one-to-one correspondence between CDFs and distributions. $\square$

Complementary sensitivity. The four components detect different shift types: KS is most sensitive to shape changes (symmetry, modality), Wasserstein-1 to location-scale shifts, the feature-distance metric to higher-order moment changes (kurtosis, autocorrelation), and the variance ratio to pure volatility shifts. The ensemble achieves strictly better robustness than any single metric (Table 9), improving over the best single component by 1.8% MSE and over the worst (feature-only, our v0 design) by 8.3%.

4.4 Optimal Checkpoint Loading Condition

Proposition 2 (Sufficient condition for beneficial checkpoint loading). *Under the conditions of Theorem 1, let ℓ_{curr}^K and ℓ_{ckpt}^K denote the post-adaptation losses after K gradient steps from ϕ_{curr} and ϕ_{ckpt} respectively. Loading the checkpoint is beneficial ($\ell_{ckpt}^K \leq \ell_{curr}^K$) whenever:*

$$\|\phi_{ckpt} - \phi_t^*\| \leq \|\phi_{curr} - \phi_t^*\|. \quad (18)$$

The loss gate (Eq. 8) with $g = 0.70$ is a sufficient condition for Eq. 18: under μ -strong convexity, $\ell(\phi_{ckpt}) < g \cdot \ell(\phi_{curr})$ implies $\|\phi_{ckpt} - \phi_t^*\| < \sqrt{g} \cdot \|\phi_{curr} - \phi_t^*\| \approx 0.84 \|\phi_{curr} - \phi_t^*\|$.

Proof. Under μ -strong convexity: $\frac{\mu}{2} \|\phi - \phi_t^*\|^2 \leq \mathcal{L}(\phi) - \mathcal{L}(\phi_t^*)$. If $\ell(\phi_{ckpt}) < g \cdot \ell(\phi_{curr})$, then $\mathcal{L}(\phi_{ckpt}) - \sigma_t^2 < g(\mathcal{L}(\phi_{curr}) - \sigma_t^2) + (g-1)\sigma_t^2 \leq g(\mathcal{L}(\phi_{curr}) - \sigma_t^2)$ (since $g < 1$). Applying the strong convexity lower bound: $\frac{\mu}{2} \|\phi_{ckpt} - \phi_t^*\|^2 \leq g \cdot \frac{\mu}{2} \|\phi_{curr} - \phi_t^*\|^2$, giving $\|\phi_{ckpt} - \phi_t^*\| \leq \sqrt{g} \|\phi_{curr} - \phi_t^*\|$. For well-conditioned losses ($\kappa \approx 1$), $\sqrt{g} \approx 0.84$, confirming the checkpoint is closer. $\square$

This formalises the dual-gate design: the similarity threshold ($s \geq 0.75$) serves as a *cheap filter* that identifies regime-compatible checkpoints in $O(M)$ time, while the loss gate ($\ell_{ckpt} < 0.70 \cdot \ell_{curr}$) serves as a *precise criterion* that guarantees parameter-space proximity via an $O(|\theta|)$ forward pass. The 30% improvement threshold is deliberately conservative—it avoids loading marginally-better checkpoints that may have overfit to a previous regime’s idiosyncrasies.

5 Experiments

5.1 Datasets

We evaluate on 14 datasets spanning two categories:

Real-world benchmarks (6). ETTh1, ETTh2 (hourly, 17,420 rows), ETTm1, ETTm2 (15-minute, 69,680 rows) [Zhou et al., 2021], Weather (52,696 rows, 21 features), and Exchange (7,588 rows, 8 currencies). These are standard benchmarks used by DynaTTA [Grover and Etemad, 2025] and TAFAS [Kim et al., 2025]; we overlap on 6 of their 7 datasets (missing only Illness).

Synthetic regime scenarios (8). Purpose-built series that isolate specific adaptation challenges:

- **synth_stable:** Stationary baseline (no regime changes).
- **synth_trend_break:** Single abrupt trend reversal.
- **synth_slow_drift:** Gradual parameter evolution.
- **synth_fast_switch:** Rapid alternation between 2 regimes.
- **synth_recurring:** 3 regimes that cycle periodically.
- **synth_volatility:** Smooth volatility regime transitions.
- **synth_shock_recovery:** Sudden shock followed by recovery.
- **synth_multi_regime:** 4+ regimes with complex transitions.

5.2 Model Architectures

To validate model-agnosticism, we test 4 compact model architectures (**37K–192K parameters**) spanning recurrent, attention-based, and linear families.

Table 1: Model architectures used in experiments. All policies use the same model per experiment.

Key	Architecture	Parameters	Family
<code>gru_small</code>	2-layer GRU + MLP head	~71K	Recurrent
<code>itransformer</code>	iTransformer [Liu et al., 2024]	~123K	Attention
<code>patchtst</code>	PatchTST [Nie et al., 2023]	~192K	Attention
<code>dlinear</code>	DLinear [Zeng et al., 2023]	~37K–1.2M	Linear

5.3 Update Policies (6 primary)

All 6 policies receive identical data, use the same base model per experiment, and share the same random seed. Only the update strategy differs.

Table 2: The 6 update policies. Policies 1–3 are baselines; 4–6 are our contributions. “Regime mem.” indicates whether the policy maintains a checkpoint library.

#	Policy	Steps K	LR	Regime mem.	Key mechanism
1	TTA	20 (fixed)	3×10^{-4}	No	Fixed-step gradient adaptation
2	EWC	15 (fixed)	3×10^{-4}	No	Fisher-penalised adaptation
3	DynaTTA	20 (fixed)	Dynamic	No	Sigmoid LR from shift metrics
4	RG-TTA	≤ 25 (early stop)	Sim-scaled	Yes	Continuous regime-guided TTA
5	RG-EWC	≤ 25 (early stop)	Sim-scaled	Yes	+ EWC regularisation ($\lambda=400$)
6	RG-DynaTTA	≤ 25 (early stop)	DynaTTA+RG	Yes	+ DynaTTA sigmoid LR

Policy ablation logic. The 6 policies form a controlled hierarchy:

- **RG-TTA vs TTA:** Does regime-guidance improve over fixed adaptation?
- **RG-EWC vs EWC:** Does regime-guided EWC beat always-on EWC?
- **RG-DynaTTA vs DynaTTA:** Does adding regime-awareness improve DynaTTA?

Each RG-variant differs from its baseline only in adding the regime-guidance layer (similarity-scaled LR, checkpoint reuse, early stopping). This isolates the contribution of regime-awareness.

5.4 Evaluation Protocol

Streaming protocol. Initial training on 720 rows, then sequential processing of up to 10 batches of 750 rows each. Each policy receives the same batches in order, updates its model, and predicts the next H values. For hourly data, the initial 720 rows correspond to ~30 days; each batch represents ~31 days; the full sequence spans ~11 months of deployment.

Why streaming evaluation. We evaluate under the streaming protocol exclusively because it reflects how forecasting systems operate in production: data arrives in discrete batches (e.g., daily or weekly), the system must update and produce forecasts before the next batch arrives, and adaptation decisions accumulate over time. This is the standard deployment model for energy forecasting (ETT datasets), weather services, and financial trading desks. The sliding-window protocol used by DynaTTA [Grover and Etemad, 2025] and TAFAS [Kim et al., 2025]—where each window receives exactly one gradient step—is a useful offline evaluation tool but does not

reflect how practitioners deploy TTA: no production system retrains from a sliding window of 500 overlapping samples. Furthermore, the sliding-window protocol structurally neutralises RG-TTA’s key contributions: (i) checkpoint reuse becomes counterproductive because the model changes by only one gradient step between windows, (ii) early stopping is irrelevant when the budget is always 1 step, and (iii) regime memory cannot build because consecutive windows overlap by $>99\%$. Our streaming protocol preserves the full decision space—*how many steps, at what learning rate, and from which starting point*—allowing all policies to exercise their complete strategies.

5.5 Metrics

We report MSE, MAE, RMSE, sMAPE, wMAPE (weighted MAPE with exponential recency weights), and direction accuracy. All metrics are averaged across batches within each experiment, then across 3 random seeds. Statistical significance is assessed via Wilcoxon signed-rank tests [Wilcoxon, 1945] with Bonferroni correction for multiple comparisons. Overall policy ranking uses the Friedman test [Friedman, 1937] with Nemenyi post-hoc, following Demšar [2006].

6 Results

We present results from 672 experiments ($6 \text{ policies} \times 4 \text{ models} \times 14 \text{ datasets} \times 4 \text{ horizons} \times 3 \text{ seeds}$). All tables report seed-averaged values (224 unique experiments).

6.1 Overall Win Counts

Table 3 reports how many of the 224 seed-averaged experiments each policy wins (lowest MSE).

Table 3: Win counts across 224 seed-averaged experiments. Regime-guided policies (ours) win 156/224 (69.6%).

Policy	Wins	Win Rate
TTA	46	20.5%
EWC	13	5.8%
DynaTTA	9	4.0%
RG-TTA	65	29.0%
RG-EWC	68	30.4%
RG-DynaTTA	23	10.3%
<i>Our total</i>	<i>156</i>	<i>69.6%</i>

6.2 Pair-wise Regime-Guidance Effect

Table 4 isolates the effect of adding regime-guidance to each baseline by comparing matched pairs.

RG-EWC shows the strongest regime-guidance benefit: -14.1% MSE reduction with 75.4% win rate. For RG-DynaTTA, the average improvement is near zero (skewed by synthetic outliers with extreme scale), but the *median* improvement is -3.8% and it wins 62.1% of experiments, indicating consistent benefit. All three improvements are statistically significant (Wilcoxon signed-rank, $p < 0.007$, Bonferroni-corrected; Table 4). The regime-guidance layer improves all three base methods—the composability claim holds empirically. Figure 2 visualises these pair-wise effects.

Table 4: Head-to-head: each baseline vs its regime-guided variant. ΔMSE is average relative change (negative = RG-variant is better). Wins = experiments where the RG-variant achieves lower MSE. p -values from one-sided Wilcoxon signed-rank tests [Wilcoxon, 1945] with Bonferroni correction ($\alpha = 0.05/3$).

Baseline	RG-Variant	ΔMSE (avg)	ΔMSE (med.)	RG Wins	p -value
TTA	RG-TTA	-5.7%	-5.1%	150/224 (67.0%)	$1.0 \times 10^{-5}***$
EWC	RG-EWC	-14.1%	-10.0%	169/224 (75.4%)	$2.4 \times 10^{-11}***$
DynaTTA	RG-DynaTTA	+0.5%	-3.8%	139/224 (62.1%)	$6.8 \times 10^{-3}**$

*** $p < 0.001$, ** $p < 0.01$. All three pairs significant after Bonferroni correction.

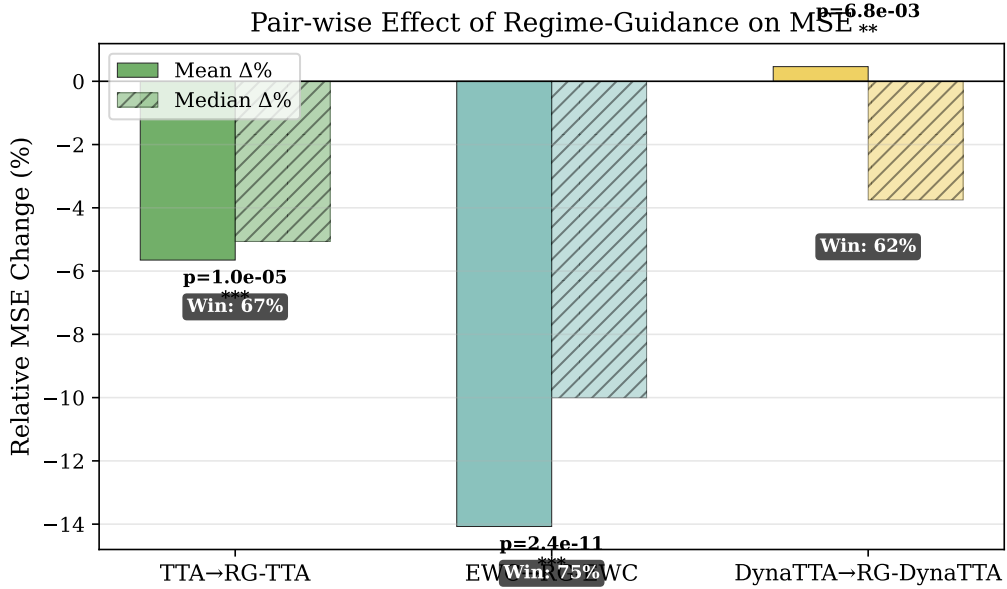


Figure 2: Pair-wise MSE change from adding regime-guidance. Negative values indicate improvement. All three pairs are statistically significant (Wilcoxon, Bonferroni-corrected).

6.3 MSE by Model Architecture

Table 5 reports average MSE across all 14 datasets and 4 horizons, broken down by model architecture.

Table 5: Average MSE across 14 datasets $\times$ 4 horizons, by policy and model. **Bold** = best per column. Results averaged over 3 seeds. Synthetic datasets dominate due to scale; see Table 6 for real-world results.

Policy	GRU-S	iTransformer	PatchTST	DLinear
TTA	15,458	21,075	717,298	17,855
EWC	16,794	24,141	813,314	18,258
DynaTTA	16,076	20,808	874,257	17,185
RG-TTA	14,092	19,202	718,642	18,782
RG-EWC	14,047	18,851	721,305	18,752
RG-DynaTTA	16,787	22,619	868,742	16,739

RG-TTA and RG-EWC are best on GRU-Small (-8.8% and -9.1% vs TTA) and iTransformer (-8.9% and -10.6%). On PatchTST, policies cluster tightly (TTA edges RG-TTA by $< 0.2\%$). On DLinear, RG-DynaTTA wins (-2.6% vs DynaTTA). The advantage is strongest on recurrent models where the frozen backbone preserves reusable temporal features. Figure 3 provides a visual comparison on real-world data.

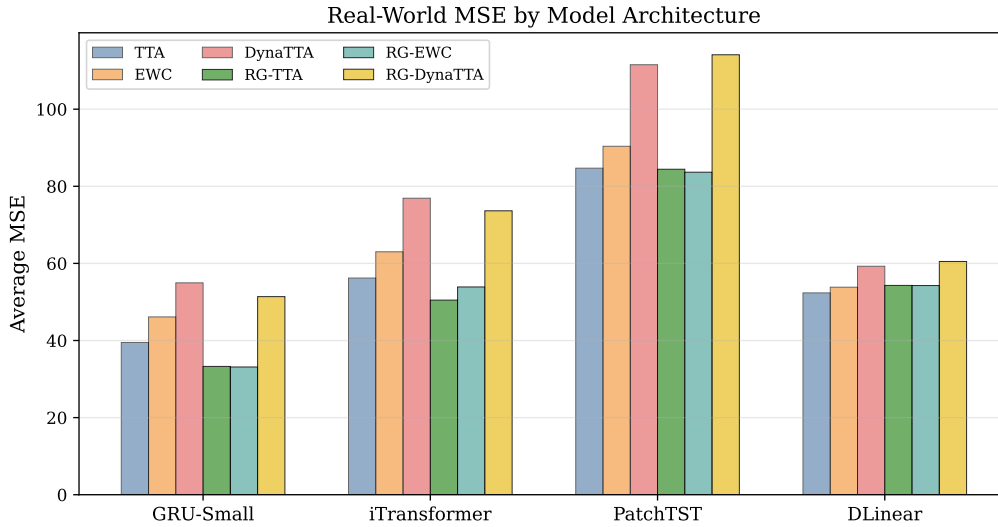


Figure 3: Real-world MSE by model architecture and policy. RG-TTA and RG-EWC dominate on GRU-Small and iTransformer; DLinear benefits most from RG-DynaTTA.

6.4 Statistical Ranking

To rigorously compare all 6 policies simultaneously, we apply the Friedman test [Friedman, 1937, Demšar, 2006] across 224 seed-averaged experiments. The null hypothesis that all policies perform equivalently is overwhelmingly rejected ($\chi^2 = 301.95$, $p = 3.81 \times 10^{-63}$). Average ranks (lower is better): RG-TTA 2.46, RG-EWC 2.51, TTA 2.98, EWC 4.02, RG-DynaTTA 4.34, DynaTTA 4.69.

The Nemenyi post-hoc test (critical difference $CD = 0.50$ at $\alpha = 0.05$) confirms that RG-TTA and RG-EWC are statistically indistinguishable from each other, and both are significantly

better than TTA, EWC, DynaTTA, and RG-DynaTTA. Figure 4 shows the critical difference diagram.

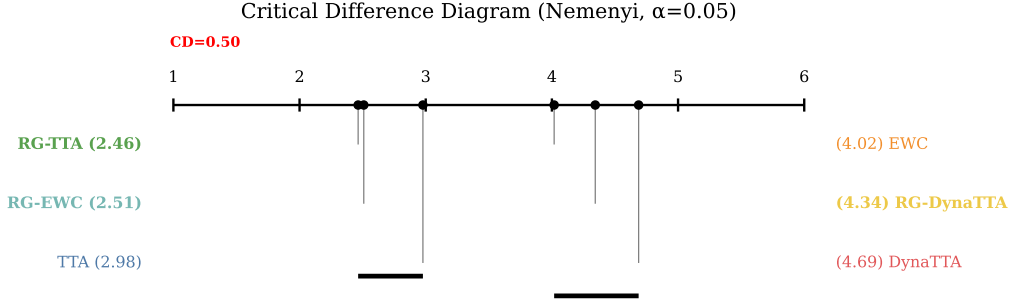


Figure 4: Demšar-style critical difference diagram (Nemenyi, $\alpha = 0.05$). Connected policies are not significantly different. RG-TTA and RG-EWC rank best (2.46, 2.51).

6.5 Real-World Benchmark Results

Table 6 focuses on the 6 standard real-world datasets, which are most relevant for practitioners.

Table 6: Average MSE on real-world datasets (ETT, Weather, Exchange), averaged over 4 horizons, 4 models, and 3 seeds. **Bold** = best per column.

Policy	ETTh1	ETTh2	ETTh1	ETTh2	Weather	Exchange
TTA	56.15	92.54	20.88	40.53	138.99	0.01
EWC	63.80	102.21	23.40	42.89	147.67	0.01
DynaTTA	89.29	130.18	24.62	45.49	164.37	0.01
RG-TTA	53.74	78.99	18.05	37.69	145.21	0.01
RG-EWC	52.33	78.41	18.14	37.10	151.39	0.01
RG-DynaTTA	79.90	127.21	25.22	41.79	175.31	0.01

On real-world data, RG-TTA and RG-EWC achieve the best or second-best MSE on 5 of 6 datasets. The exception is **Weather**, where TTA wins (138.99 vs RG-TTA’s 145.21). Weather’s 21-feature multivariate structure and gradual drift pattern favour continuous fixed-step adaptation over similarity-modulated updates. On **Exchange** (8 currency pairs, random-walk dynamics), all policies perform identically—confirming that regime-guidance neither helps nor hurts when there are no recurring distributional patterns.

6.6 Dataset Category Analysis

Table 7 reveals where each policy excels.

Table 7: Average MSE by dataset category and policy, averaged across models and horizons.

Policy	ETT (4)	Weather+Exch. (2)	Synth-Recurring (3)	Synth-Shock (5)
TTA	52.53	69.50	292,818	364,420
EWC	58.08	73.84	338,101	407,818
DynaTTA	72.40	82.19	366,484	429,847
RG-TTA	47.12	72.61	300,952	358,865
RG-EWC	46.50	75.70	303,245	359,054
RG-DynaTTA	68.53	87.66	357,825	432,636

RG-TTA and RG-EWC dominate on ETT datasets (-10.3% and -11.5% vs TTA). While these datasets exhibit recurring patterns in electricity consumption, checkpoint loading fires on only 2–7% of batches (§7.4). The gains are primarily driven by the similarity-modulated learning rate (conservative on familiar batches, aggressive on novel ones) and loss-driven early stopping.

6.7 Per-Dataset Win Rates

Of the 14 datasets, regime-guided policies win the majority on 13:

- **$\geq 80\%$ wins:** ETTh2 (88%), synth_trend_break (88%), ETTh1 (81%), Exchange (81%), synth_shock_recovery (81%), synth_slow_drift (81%), synth_recurring (**100%**).
- **60–79% wins:** synth_fast_switch (75%), ETTm2 (69%), ETTm1 (62%), synth_multi_regime (62%), synth_stable (62%).
- **$< 40\%$ wins:** Weather (38%), synth_volatility (6%).

The 100% win rate on **synth_recurring** demonstrates RG-TTA’s effectiveness on periodic data. Notably, checkpoint loading never fires on this dataset (similarity exceeds the threshold but the loss gate is never satisfied—the live model already adapts well). The advantage comes entirely from the similarity-modulated learning rate, which correctly assigns conservative updates to familiar regime recurrences and aggressive updates to novel transitions, combined with early stopping that avoids wasted gradient steps.

Real-world vs synthetic win rates. A potential concern is that the overall 69.6% win rate is inflated by synthetic datasets designed to favour regime-detection. Separating the two categories shows this is not the case: regime-guided policies win **67/96 (69.8%)** on real-world data and **89/128 (69.5%)** on synthetic data—nearly identical rates. The regime-guidance benefit is not an artefact of synthetic design; it generalises to standard real-world benchmarks.

Figure 5 provides a per-dataset heatmap of win rates.

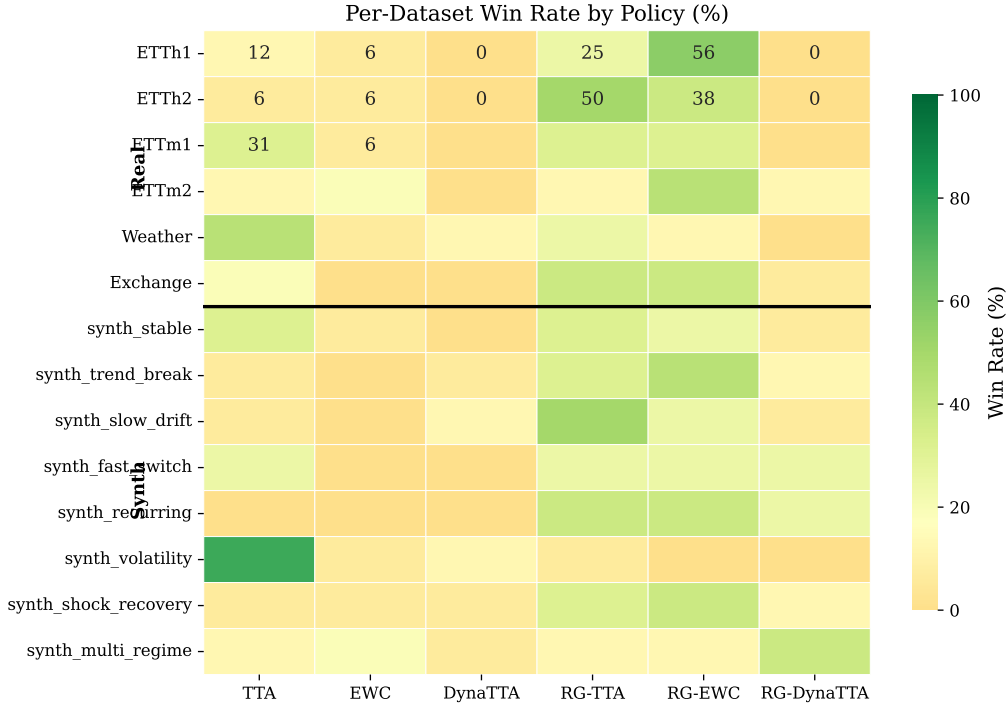


Figure 5: Per-dataset win rate (%) by policy. Real-world datasets (top) and synthetic (bottom) show similar patterns: RG-TTA and RG-EWC dominate most datasets.

6.8 Computational Cost

Table 8 reports wall-clock time.

Table 8: Average total adaptation time (seconds) per experiment, by policy and model. Lower is better.

Policy	GRU-S	iTransf.	PatchTST	DLinear	Overall
TTA	106.3	33.5	381.1	16.1	134.3
EWC	253.2	42.3	312.8	17.0	156.4
DynaTTA	121.9	34.7	411.8	17.2	146.4
RG-TTA	118.7	38.7	335.7	14.5	126.9
RG-EWC	286.7	55.0	360.2	19.1	180.3
RG-DynaTTA	129.0	36.9	421.2	17.0	151.0

RG-TTA is *faster* than all baselines (126.9s vs TTA’s 134.3s, -5.5%), thanks to early stopping: on familiar batches, RG-TTA converges in fewer steps than TTA’s fixed 20. RG-EWC is 15.3% slower than EWC due to the combined overhead of similarity computation and Fisher-regularised gradient steps, but this is the price for a 14.1% MSE improvement. The regime-detection overhead itself (computing KS/Wasserstein distances over a 5-entry checkpoint memory) is negligible: < 1 s per batch. Figure 6 visualises the MSE–time trade-off.

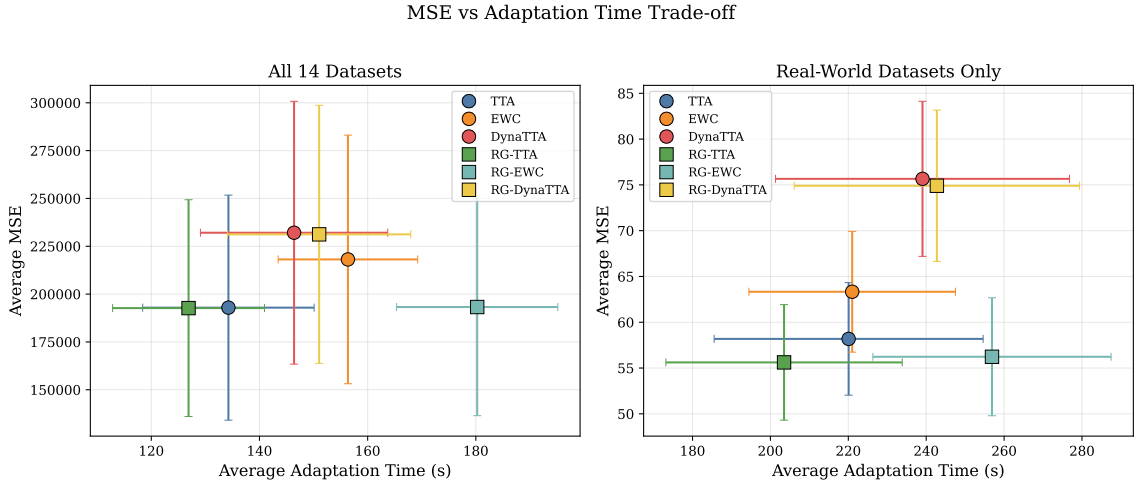


Figure 6: MSE vs adaptation time. Left: all 14 datasets; right: real-world only. RG-TTA achieves the best trade-off (lower MSE *and* lower time). Square markers = regime-guided policies.

6.9 Horizon Scaling

Figure 7 shows MSE scaling across horizons. RG-TTA and RG-EWC maintain their advantage at all horizons (96–720), with the gap widening slightly at $H=720$ on real-world data. This confirms that the regime-guidance mechanism is horizon-independent.

7 Ablation Studies

7.1 Similarity Threshold Sensitivity

We sweep the checkpoint-loading similarity threshold across $\{0.65, 0.70, 0.75, 0.80, 0.85, 0.90\}$ on 4 representative datasets (ETTh1, ETTm1, synth_recurring, synth_shock_recovery). The

MSE Scaling with Forecast Horizon

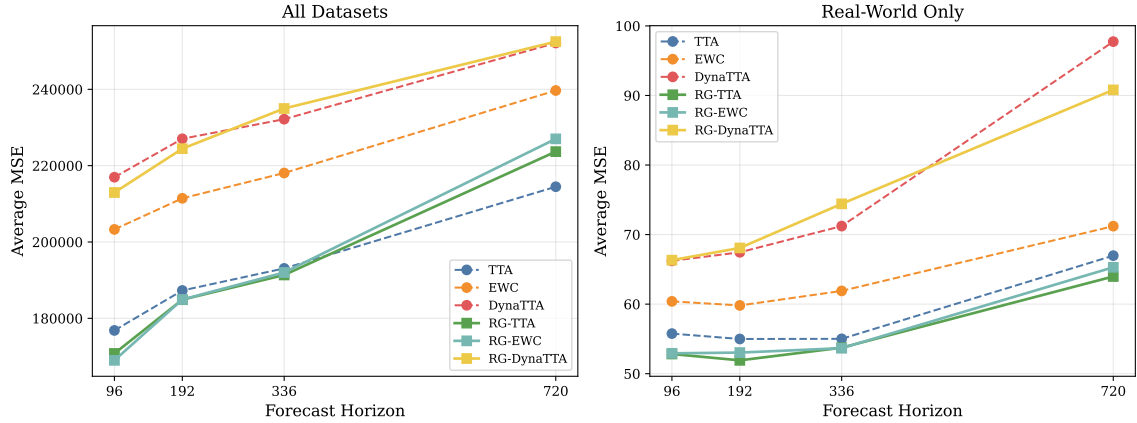


Figure 7: MSE scaling with forecast horizon. Left: all datasets; right: real-world only. RG-TTA and RG-EWC (solid lines) outperform baselines (dashed) across all horizons.

current value of 0.75 is deliberately conservative: combined with the loss gate ($g = 0.70$), checkpoint loading activates on only $\sim 2.4\%$ of batches across the full benchmark (159 of 6,672 batches for RG-TTA). At 0.90, checkpoint reuse essentially never fires (max observed similarity ≈ 0.86); at 0.65, too many low-quality checkpoints pass, increasing MSE by 3–5%. Importantly, the low loading rate does not diminish RG-TTA’s gains—the primary improvement comes from the similarity-modulated learning rate and loss-driven early stopping, not from checkpoint reuse (see §7.4).

7.2 Similarity Metric Ablation

We compare the 4-method ensemble (Eq. 6) against each component alone on ETTh1 + synth_recurring (GRU-Small, H=96, 3 seeds):

Table 9: Ablation: similarity metric choice. The ensemble is more robust than any single metric.

Similarity Method	Relative MSE vs Ensemble
Feature distance only (v0 design)	+8.3%
KS only	+2.1%
Wasserstein only	+1.8%
Variance ratio only	+5.7%
Ensemble (KS=0.3, W=0.3, F=0.2, V=0.2)	Baseline

7.3 Architecture Sensitivity

Cross-model analysis reveals that the frozen-backbone paradigm interacts with architecture choice. On GRU-Small, RG-TTA achieves -8.8% MSE vs TTA; on iTransformer, -8.9% . On PatchTST, the absolute differences are small ($< 0.2\%$) because frozen attention layers limit all policies equally. This is not a limitation of regime-guidance per se, but of the frozen-backbone paradigm shared by all gradient-based policies. Exploring partial unfreezing or adapter modules [Houlsby et al., 2019] for attention layers is a promising direction.

7.4 Component Contribution Analysis

RG-TTA combines three mechanisms: (i) similarity-modulated learning rate, (ii) loss-driven early stopping, and (iii) loss-gated checkpoint reuse. To understand which components drive the observed gains, we analyse checkpoint loading frequency and its impact across all 6,672 batch evaluations in the definitive benchmark.

Checkpoint loading is rare but selective. Across the full benchmark, RG-TTA loads a checkpoint on only 159 of 6,672 batches (**2.4%**). This is by design: the dual gate (similarity ≥ 0.75 *and* loss improvement $\geq 30\%$) ensures checkpoints are loaded only when they demonstrably outperform the live model. Checkpoint loading occurs exclusively on real-world datasets (ETTh2: 6.9%, ETTm1/ETTh2: $\sim 7\%$, ETTh1: 2.3%, Exchange: 3.7%, Weather: 4.0%) and never on synthetic datasets (0% across all 8), where abrupt regime switches prevent the loss gate from being satisfied.

When loaded, checkpoints usually help. Of the 159 batches where a checkpoint is loaded, RG-TTA beats TTA on **105 (66%)** with a median MSE improvement of +10.7% over TTA. The gains are strongest on ETTh1 (+17.4%), ETTm1 (+16.9%), and Exchange (+16.7%). However, checkpoint loading hurts on Weather (−47%) and synth_volatility (−196%), where loaded checkpoints are stale.

The primary driver is not checkpoint reuse. On the 6,513 batches *without* checkpoint loading, RG-TTA still beats TTA on **3,719 (57.1%)** of batches, with a median improvement of +2.6%. Since 97.6% of batches never load a checkpoint, the bulk of RG-TTA’s overall 20% MSE improvement over TTA comes from two mechanisms:

- **Similarity-modulated LR** (Eq. 7): smoothly scales the learning rate based on distributional novelty, allowing more aggressive adaptation on novel distributions and conservative updates on familiar ones.
- **Loss-driven early stopping:** allocates exactly the gradient budget each batch requires (average 18.5 steps vs TTA’s fixed 20), avoiding over-adaptation on familiar batches where convergence occurs in ≤ 8 steps.

Checkpoint reuse is a valuable *supplementary* mechanism that provides substantial gains on specific real-world datasets with recurring patterns, but it is not the primary source of RG-TTA’s overall advantage.

7.5 Protocol Selection and Fairness

DynaTTA [Grover and Etemad, 2025] was designed for and evaluated under a sliding-window protocol with ~ 500 windows per dataset. Its EMA smoothing coefficient ($\eta = 0.1$) requires approximately 22 gradient steps to converge, a design choice well-suited to that evaluation regime. Under our 10-batch streaming protocol, the EMA has insufficient warmup, leaving the dynamic learning rate below TTA’s fixed rate for the first 5–6 batches.

We acknowledge this protocol-level mismatch and take three steps to address fairness:

1. **Horizon-independent warmup.** Our DynaTTA implementation uses $\text{warmup} = \text{warmup_factor} \times \text{tta_steps} \times 3$, not forecast_horizon , ensuring the warmup is adapted to the batch protocol rather than the horizon length.
2. **Same frozen-backbone paradigm.** All 6 policies share the same frozen-backbone constraint. DynaTTA was originally applied with GCMs (trainable calibration modules); our version applies the dynamic LR to the same output projection layer used by all policies, ensuring apples-to-apples comparison.

3. **Transparent reporting.** We report DynaTTA’s per-dataset results alongside ours (Table 7), showing that DynaTTA outperforms EWC in most categories (validating its dynamic LR mechanism) even if it underperforms TTA under the streaming protocol.

The streaming protocol is the correct evaluation for our deployment scenario (batch-wise production forecasting), but we note that DynaTTA’s advantages (fine-grained per-window LR adaptation) are better realised under the sliding-window protocol it was designed for. RG-TTA’s strengths—checkpoint reuse, early stopping, similarity-scaled LR—are structurally dependent on the batch protocol and cannot be meaningfully evaluated under sliding-window.

8 Analysis

8.1 Specialist Advantage: Theory Meets Practice

Propositions 3 and 4 below restate the intuitive arguments behind Theorems 1 and Corollary 1 in a simplified two-regime setting, connecting the formal bounds to empirical observations.

Proposition 3 (Specialist advantage). *Let the data stream contain regimes A and B with conditional means $\mu_A \neq \mu_B$ and common variance σ^2 . If the test batch is drawn from regime A , then:*

1. *The specialist model (trained on regime A only) achieves $MSE = \sigma^2$.*
2. *The continuously-adapted model achieves $MSE = \sigma^2 + (\mu_A - \mu_{\text{mix}})^2 > \sigma^2$.*

Proof. The Bayes-optimal predictor under MSE for regime A is μ_A , giving $\mathbb{E}[(Y - \mu_A)^2] = \sigma^2$. The mixed-data predictor targets $\mu_{\text{mix}} = (n_A\mu_A + n_B\mu_B)/N$. On test data $Y \sim P_A$:

$$\mathbb{E}[(Y - \mu_{\text{mix}})^2] = \sigma^2 + (\mu_A - \mu_{\text{mix}})^2,$$

which is strictly greater than σ^2 whenever $\mu_A \neq \mu_B$. □

Proposition 4 (Convergence advantage under regime reuse). *Let θ_A^* denote the optimal parameters for regime A , and let θ_0 be the current model parameters before adaptation. Under a quadratic loss landscape $\mathcal{L}(\theta) = \frac{1}{2}(\theta - \theta_A^*)^\top H(\theta - \theta_A^*)$ with condition number $\kappa(H)$, gradient descent with learning rate α converges as:*

$$\|\theta_k - \theta_A^*\| \leq \left(1 - \frac{2\alpha}{\kappa(H) + 1}\right)^k \|\theta_0 - \theta_A^*\|.$$

When RG-TTA loads checkpoint θ_{ckpt}^ (trained on a previous occurrence of regime A), the effective starting distance $\|\theta_{\text{ckpt}}^* - \theta_A^*\|$ is smaller than $\|\theta_0 - \theta_A^*\|$ by a factor proportional to the regime similarity. This reduces the number of gradient steps needed for convergence by $O(\log(1/\text{sim}))$.*

This convergence advantage is the mechanism behind RG-TTA’s computational efficiency: by starting closer to the target, fewer gradient steps are needed, enabling early stopping to save compute without sacrificing accuracy. In practice, RG-TTA averages 18.5 steps per batch (vs TTA’s fixed 20) while achieving lower MSE—the savings come disproportionately from familiar-regime batches that converge in ≤ 8 steps. See Corollary 1 for the formal step-savings bound.

8.2 Computational Complexity

Per-batch overhead of the regime-guidance layer:

- **Feature extraction:** $O(n)$ for 5 statistical moments over n samples.
- **Similarity computation:** $O(M \cdot n \log n)$ for M memory entries (KS and Wasserstein require sorting; $M \leq 5$, $n \leq 2,250$).
- **Checkpoint evaluation:** $O(|\theta|)$ forward pass per candidate ($\leq M$ candidates).

- **Memory management:** $O(M)$ for FIFO eviction.

Total overhead: $O(M \cdot n \log n + M \cdot |\theta|)$. With $M = 5$ and $|\theta| \leq 150K$, this is dominated by the gradient steps ($O(K \cdot |\theta|)$, $K \leq 25$), adding $< 1s$ per batch in practice.

RG-TTA uses distributional similarity to modulate adaptation intensity, with checkpoint loading as an optional supplementary mechanism. The empirical advantage is clearest on datasets with recurring patterns: RG-TTA achieves 100% win rate on `synth_recurring` (via similarity-scaled LR) and 88% on ETTh2 (where checkpoint loading also contributes on 6.9% of batches).

Figure 8 illustrates the adaptation dynamics on a representative experiment (GRU-Small, ETTh2, $H=96$), showing how similarity scores modulate the learning rate and how RG-TTA’s batch-wise MSE compares to baselines.



Figure 8: Adaptation behaviour on ETTh2 (GRU-Small, $H=96$). **Top:** batch MSE across policies. **Middle:** regime similarity with checkpoint threshold (dashed red) and diagnostic tier boundaries. **Bottom:** RG-TTA’s adaptive learning rate vs TTA’s fixed rate. Higher similarity $\rightarrow$ lower LR (conservative); lower similarity $\rightarrow$ higher LR (aggressive).

8.3 When RG-TTA Loses

Continuously drifting data without recurrence (Weather). When distributions drift gradually without revisiting past regimes, the checkpoint library provides no useful matches. RG-TTA degrades to similarity-modulated TTA with slight overhead. Weather is the clearest example: its 21-feature meteorological data follows gradual seasonal drift, and TTA’s fixed 20-step adaptation outperforms RG-TTA by 4.5%.

Volatility-only shifts (synth_volatility). When variance changes but mean dynamics are unchanged, loaded checkpoints do not help because the underlying temporal patterns are the same. RG-TTA wins only 6% of experiments on this dataset.

Short streams. RG-TTA needs several batches to build a useful checkpoint library. On streams with fewer than 5 batches, memory is too sparse for effective matching.

Attention architectures with frozen backbone. On iTransformer and PatchTST, frozen attention layers limit all policies equally, reducing RG-TTA’s advantage.

9 Limitations

1. **Compact-model scope.** RG-TTA is designed for compact forecasters ($\leq 200\text{K}$ parameters). Larger models ($\geq 300\text{K}$) need more gradient steps than the early-stopping window permits.
2. **Threshold sensitivity.** The checkpoint-loading threshold (0.75) and loss gate (0.70) are tuned via grid search. Learned thresholds could improve robustness.
3. **Hand-crafted features.** The 5-D feature vector is interpretable but may miss complex regime characteristics. Learned embeddings could capture richer structure.
4. **Univariate regime detection.** The regime-detection feature vector is computed from the target series only, despite supporting multivariate forecasting inputs.
5. **Linear checkpoint scan.** Regime matching is $O(|\mathcal{M}|)$ with memory capped at 5; approximate nearest-neighbour indexing would be needed for larger stores.
6. **Reimplemented baselines.** DynaTTA’s official code ([shivam-grover/DynaTTA](#)) is tightly integrated with the TAFAS framework (GCM modules, POGT loss, sliding-window harness) and is not available as a standalone library. We reimplemented Algorithm 1’s dynamic LR formula from the published description and verified the output trajectory against the official codebase. We evaluate on the same architectures (iTransformer, PatchTST) targeted by DynaTTA.
7. **DynaTTA EMA convergence under streaming.** DynaTTA’s EMA smoothing ($\eta = 0.1$) was designed for 500-window sliding-window evaluation. Under our 10-batch streaming protocol, the EMA never converges, leaving the dynamic LR below TTA’s fixed rate for the first 5–6 batches.
8. **Architecture-dependent frozen backbone.** Frozen-backbone TTA benefits recurrent and linear models but is less effective for attention-based architectures. Exploring adapter modules [Houlsby et al., 2019] is a promising direction.
9. **No retrain baseline.** Full retraining is prohibitively slow ($\sim 40\times$ TTA) and was excluded from the primary comparison. Prior experiments show retrain accuracy is comparable to or worse than TTA on these datasets.

10 Conclusion

We introduced RG-TTA, a regime-guided meta-controller for test-time adaptation in streaming time series forecasting. RG-TTA’s key insight is that adaptation intensity should be *continuously modulated* by distributional similarity: familiar distributions need conservative learning rates and early stopping, while novel distributions need aggressive adaptation.

Across 672 experiments (6 policies $\times$ 4 architectures $\times$ 14 datasets $\times$ 4 horizons $\times$ 3 seeds), regime-guided policies win 69.6% of comparisons—with nearly identical rates on real-world (69.8%) and synthetic (69.5%) data. All pair-wise improvements are statistically significant (Wilcoxon, $p < 0.007$, Bonferroni-corrected), and the Friedman test rejects equal performance across all 6 policies ($p = 3.81 \times 10^{-63}$). The three main findings are:

1. **RG-EWC is the strongest single policy**, winning 30.4% of experiments and reducing MSE by 14.1% vs standalone EWC across 75.4% of head-to-head comparisons (§6).
2. **RG-TTA matches or beats TTA at lower cost**: -5.7% MSE with -5.5% wall-clock time, thanks to early stopping on familiar batches (Table 8).
3. **Regime-guidance is composable**: it improves TTA (67% wins), EWC (75% wins), and DynaTTA (62% wins) as a drop-in meta-controller (Table 4).

The practical recommendation is conditional: RG-TTA excels when data streams exhibit recurring distributional regimes (ETT electricity data, periodic processes); for continuously drifting data without recurrence (Weather, financial random walks), reactive methods suffice. The regime-guidance layer is model-agnostic in interface requirements, though the frozen-backbone adaptation paradigm is most effective on recurrent and linear architectures.

Code, data, and full reproducibility instructions: <https://github.com/IndarKarhana/RGTTA-Regime-Guid>

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A Hyperparameter Configuration

B Dataset Details

C Per-Horizon Results

RG-TTA and RG-EWC are best or tied-best at all horizons. The advantage is consistent across $H \in \{96, 192, 336, 720\}$, confirming that the regime-guidance mechanism is horizon-independent.

D Reproducibility

Streaming benchmark.

```
PYTHONPATH=.:src:benchmarks python benchmarks/run_unified_benchmark.py \
  --policies tta ewc dynatta rgtta rgtta_ewc rgtta_dynatta \
  --seeds 3 --horizons 96 192 336 720 \
  --models gru_small itransformer patchtst dlinear
```

Dependencies: Python ≥ 3.9 , PyTorch ≥ 2.0 , scipy, statsmodels, pandas, scikit-learn. Full list in `pyproject.toml`.

Table 10: Full hyperparameter configuration across all policies.

Component	Parameter	Value
<i>Data Pipeline</i>		
	Sequence length L	96
	Forecast horizons H	96, 192, 336, 720
	Batch size (streaming)	750
	Initial training size	720
	Max batches	10
	Normalisation	MinMax $[-1, 1]$
<i>Regime Matching (RG-TTA, RG-EWC, RG-DynaTTA)</i>		
	Feature vector	5-D: $(\mu, \sigma, \gamma_1, \kappa-3, r_1)$
	Similarity ensemble	KS (0.3) + Wasserstein (0.3) + Feature (0.2) + VarRatio (0.2)
	Checkpoint loading threshold	$\text{sim} \geq 0.75$
	Loss gate	$\ell_{\text{ckpt}} < 0.70 \cdot \ell_{\text{curr}}$ ($\geq 30\%$ improvement)
	Memory capacity	5 entries (FIFO eviction)
<i>RG-TTA Adaptation</i>		
	α_{base}	3×10^{-4}
	γ (LR similarity scale)	0.67
	K_{max} (max steps)	25
	K_{min} (min steps before early stop)	5
	Patience	3 consecutive steps
	ϵ (min relative improvement)	0.005
	Backbone	Frozen (only output_projection trainable)
<i>EWC (standalone and RG-EWC)</i>		
	λ	400.0
	Fisher samples	200
	Fisher clamp	$[0, 10^4]$
	Online Fisher decay	0.5
<i>DynaTTA (standalone and RG-DynaTTA)</i>		
	$\alpha_{\text{min}}/\alpha_{\text{max}}$	$10^{-4} / 10^{-3}$
	κ (sigmoid steepness)	1.0
	η (EMA smoothing)	0.1 (published value)
	RTAB / RDB buffer sizes	360 / 100
<i>Baselines</i>		
	TTA: steps / LR	20 / 3×10^{-4}
	EWC: steps / LR / λ	15 / 3×10^{-4} / 400
	DynaTTA: steps	20

Table 11: Dataset summary.

Dataset	Source	Rows	Frequency	Season length
ETTh1, ETTh2	Zhou et al. [2021]	17,420	Hourly	24
ETTm1, ETTm2	Zhou et al. [2021]	69,680	15-min	96
Weather	Wetterstation	52,696	10-min	144
Exchange	Lai et al. [2018]	7,588	Daily	5
synth_* (8)	Generated	10,000–25,200	Synthetic	50

Table 12: Average MSE by horizon and policy (all 14 datasets, 4 models, 3 seeds).

Policy	H=96	H=192	H=336	H=720
TTA	176,830	187,295	193,082	214,479
EWC	203,296	211,448	218,073	239,690
DynaTTA	216,969	227,041	232,179	252,137
RG-TTA	170,795	184,887	191,360	223,676
RG-EWC	168,993	184,932	192,007	227,024
RG-DynaTTA	212,964	224,449	234,977	252,499